RotorCalc - light orange theme

Screens as live on rotorcalc.vercel.app on 13 Oct 2026. Code kept on the branch light-orange.

21 screens

RotorCalc

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IAPWS-IF97 steam properties, checked against published values

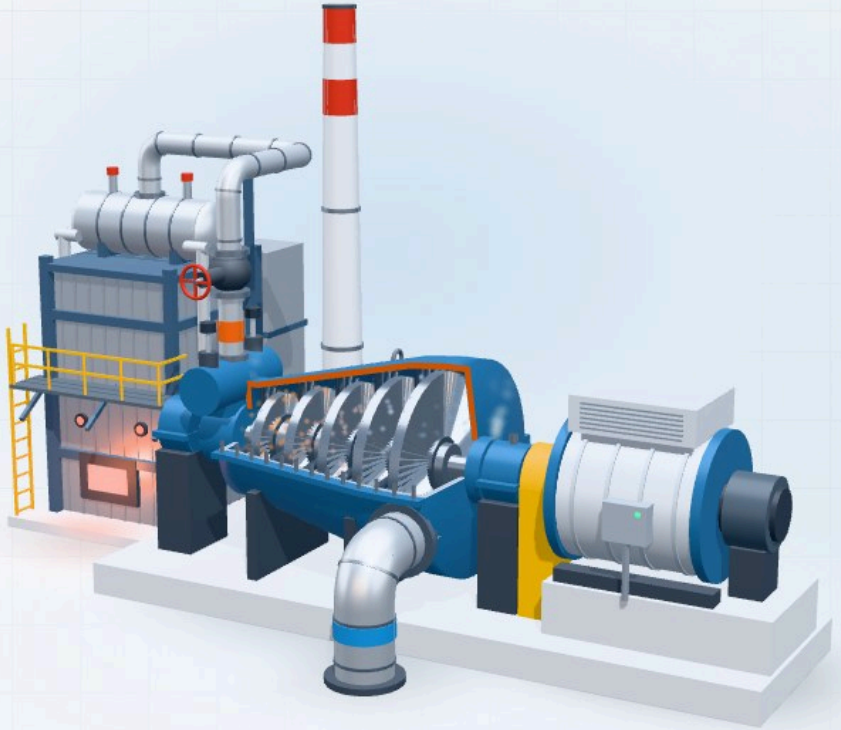
Know what your turbine should make.

Enter steam conditions. Get power output, efficiency and steam demand in seconds, with wet exhaust handled exactly and every part of the machine shown as you type.

Start free for 14 days

Try it right here ↓

No card needed. Works in any pressure, temperature and flow unit.



Drag to rotate · scroll to lift the casing

Steam flow 50 TPH

Inlet pressure 66 kg/cm²(a)

GENERATOR OUTPUT

6.73 MW

Landing page, casing lifted on scroll 2 / 21

RotorCalc

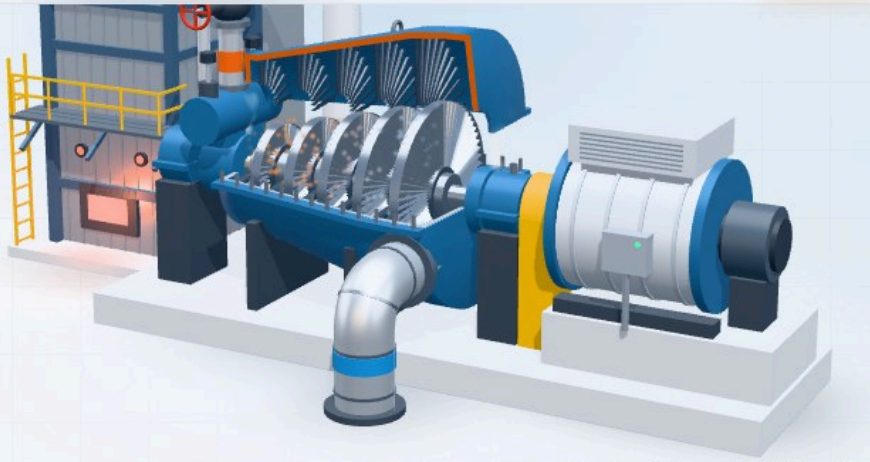
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make.

Enter steam conditions. Get power output, efficiency and steam demand in seconds, with wet exhaust handled exactly and every part of the machine shown as you type.

Start free for 14 days Try it right here ↓

No card needed. Works in any pressure, temperature and flow unit.



Drag to rotate · scroll to lift the casing

Steam flow 50 TPH

Inlet pressure 66 kg/cm²(a)

Inlet temperature 485 °C

Back pressure 3.5 kg/cm²(a)

GENERATOR OUTPUT

6.73 MW

Enthalpy drop	Steam rate	Ideal exhaust
713 kJ/kg	7.43 kg/kWh	97.1 % dry

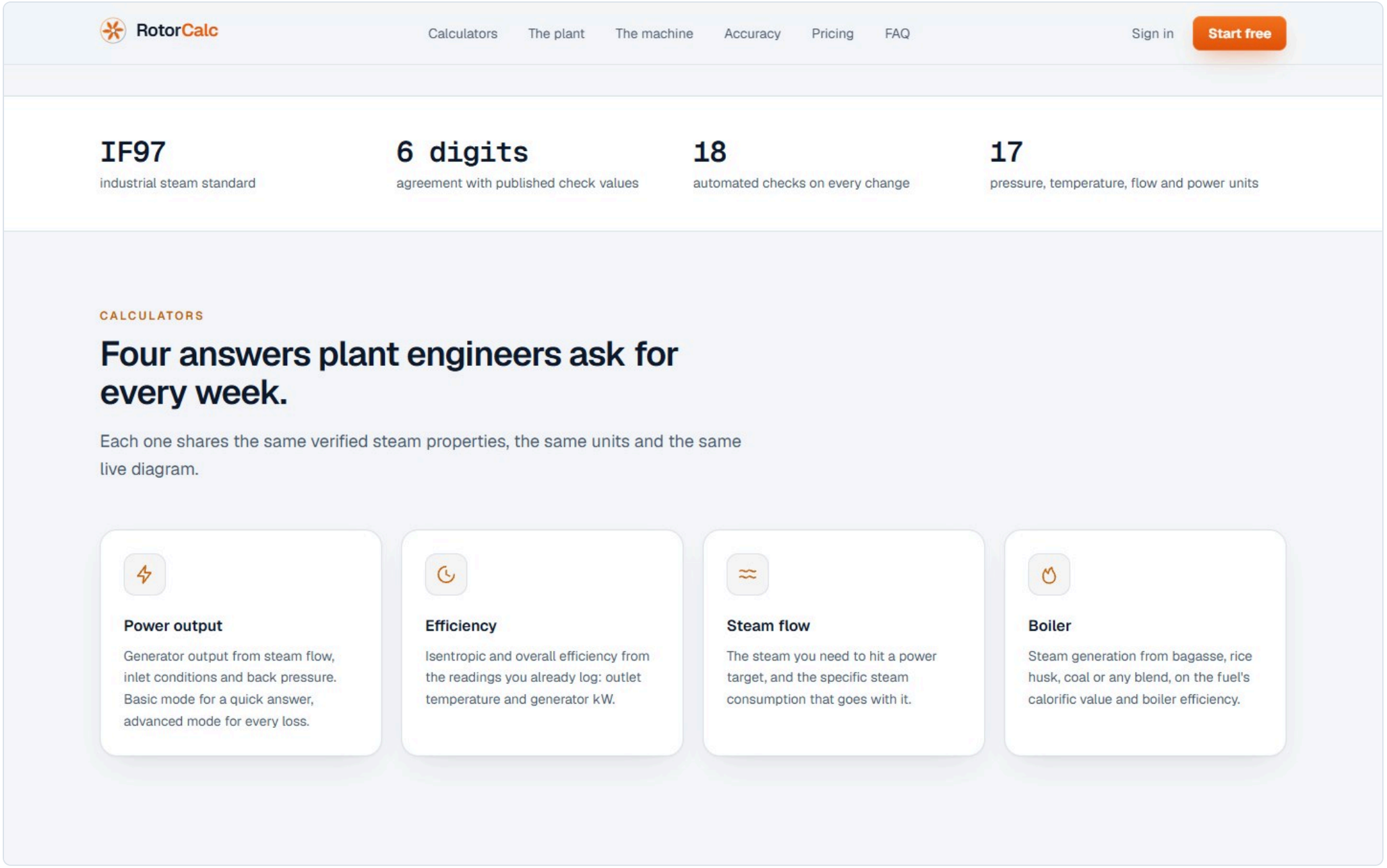
Live demo at 68 % overall efficiency. The full calculator lets you set every loss.

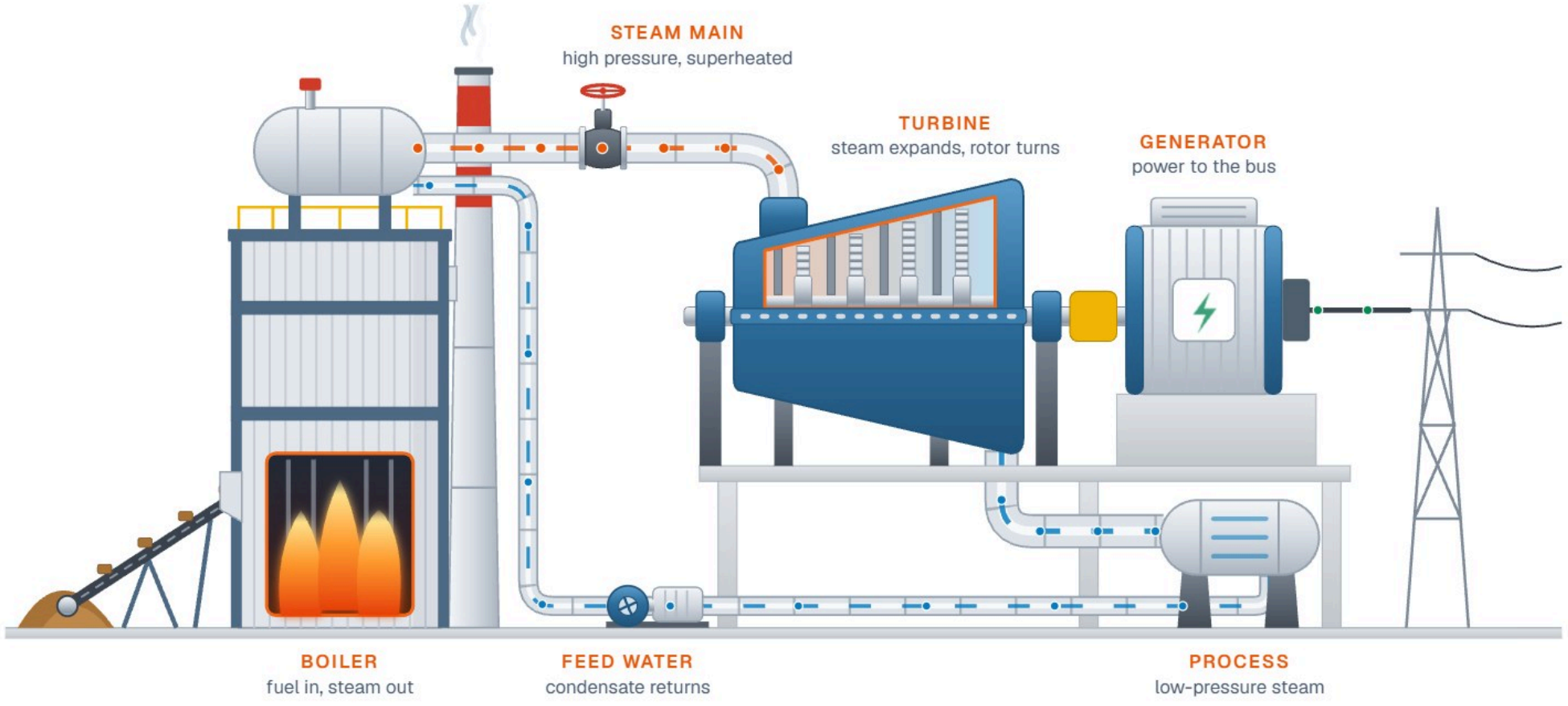
IF97
industrial steam standard

6 digits
agreement with published check values

18
automated checks on every change

17
pressure, temperature, flow and power units


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IF97
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agreement with published check values
18
automated checks on every change
17
pressure, temperature, flow and power units
CALCULATORS
Four answers plant engineers ask for every week.
Each one shares the same verified steam properties, the same units and the same live diagram.
Power output
Generator output from steam flow, inlet conditions and back pressure. Basic mode for a quick answer, advanced mode for every loss.
Efficiency
Isentropic and overall efficiency from the readings you already log: outlet temperature and generator kW.
Steam flow
The steam you need to hit a power target, and the specific steam consumption that goes with it.
Boiler
Steam generation from bagasse, rice husk, coal or any blend, on the fuel's calorific value and boiler efficiency.



Boiler
Steam generated from your fuel mix

Steam flow
Steam the turbine needs for a target

Power output
What the generator should deliver

Efficiency
How well the machine is really doing

THE MACHINE

Every input has a place on the turbine.

Point at a part to see what it does and which numbers belong to it. The calculator form works the same way.

01 Steam inlet

02 Inlet valve

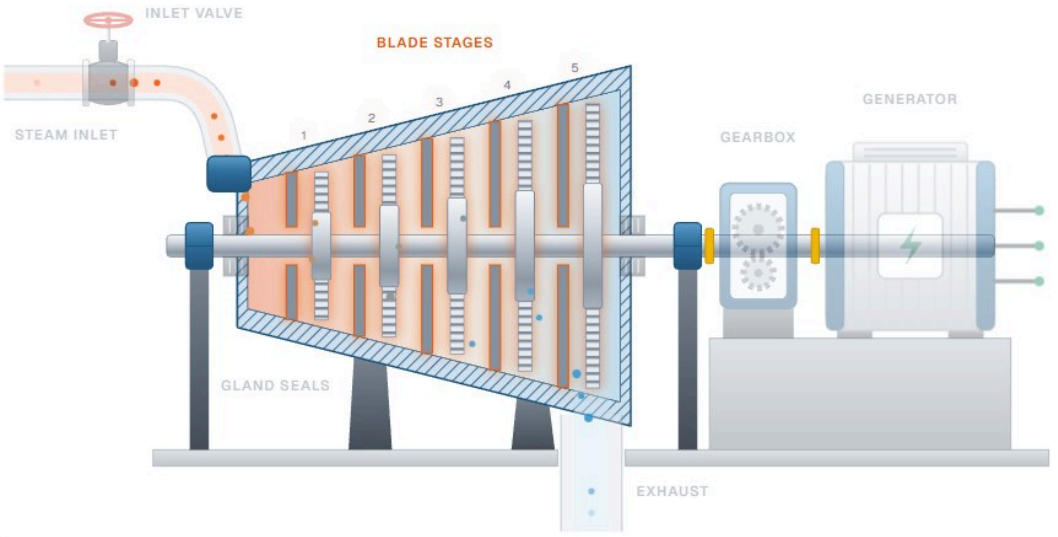
03 Blade stages

04 Gland seals

05 Exhaust

06 Gearbox

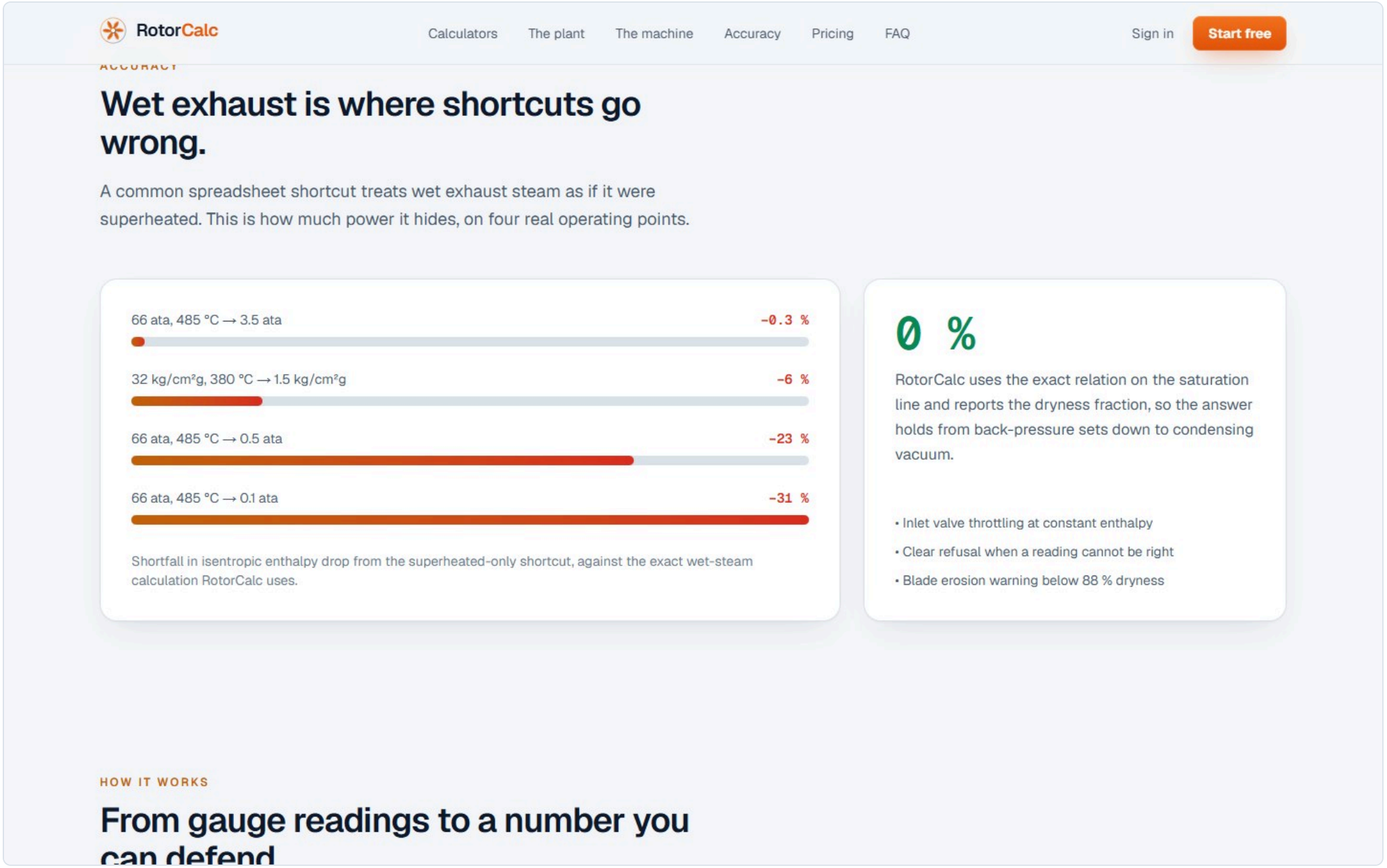
07 Generator

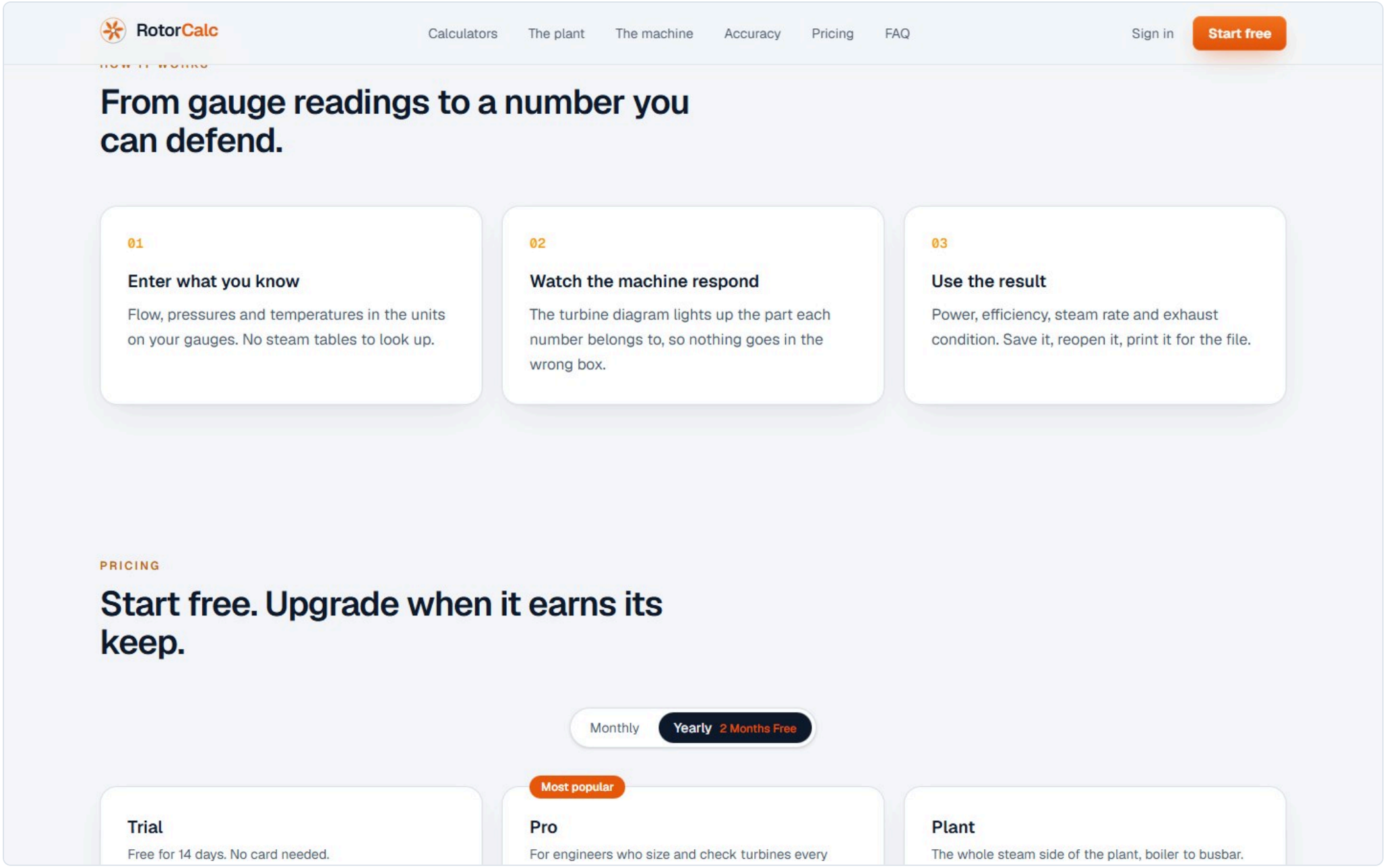


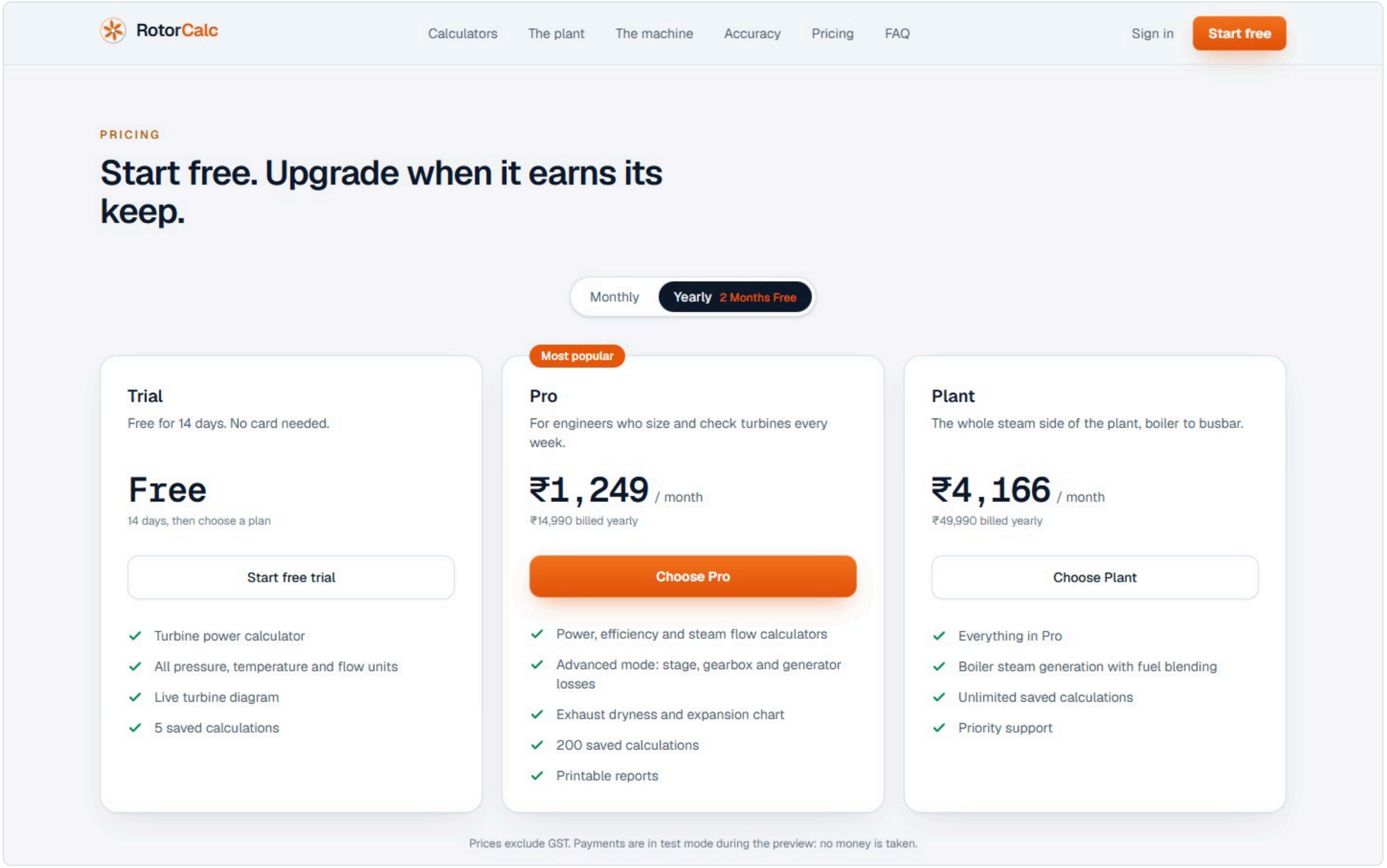
Blade stages

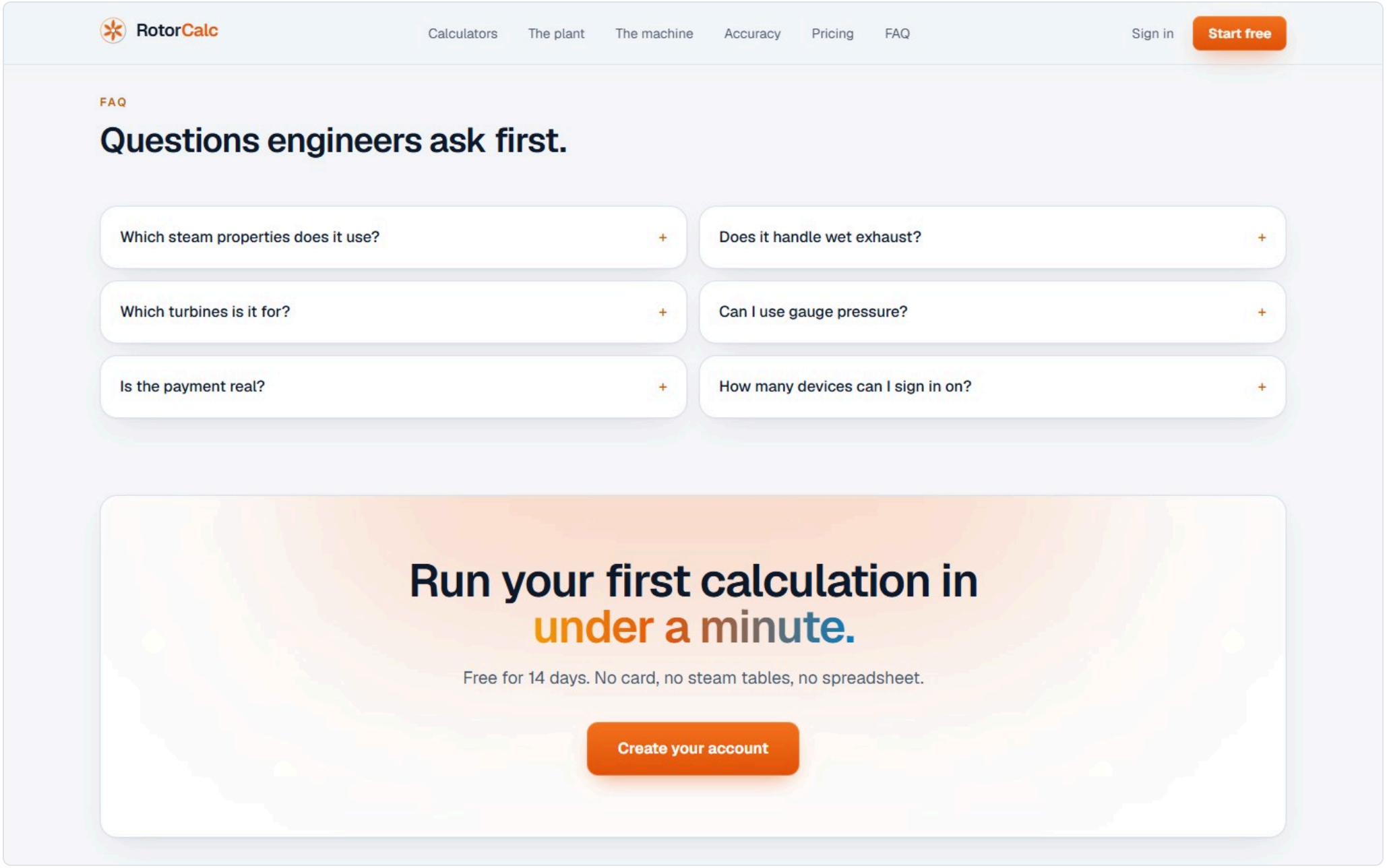
Steam expands and turns the rotor. Isentropic efficiency says how much of the ideal drop the blades capture.

You enter: Turbine type · isentropic efficiency


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ACCURACY
Wet exhaust is where shortcuts go wrong.
A common spreadsheet shortcut treats wet exhaust steam as if it were superheated. This is how much power it hides, on four real operating points.
66 ata, 485 °C → 3.5 ata
-0.3 %
32 kg/cm²g, 380 °C → 1.5 kg/cm²g
-6 %
66 ata, 485 °C → 0.5 ata
-23 %
66 ata, 485 °C → 0.1 ata
-31 %
Shortfall in isentropic enthalpy drop from the superheated-only shortcut, against the exact wet-steam calculation RotorCalc uses.
0 %
RotorCalc uses the exact relation on the saturation line and reports the dryness fraction, so the answer holds from back-pressure sets down to condensing vacuum.
• Inlet valve throttling at constant enthalpy
• Clear refusal when a reading cannot be right
• Blade erosion warning below 88 % dryness
HOW IT WORKS
From gauge readings to a number you can defend


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From gauge readings to a number you can defend.
01
Enter what you know
Flow, pressures and temperatures in the units on your gauges. No steam tables to look up.
02
Watch the machine respond
The turbine diagram lights up the part each number belongs to, so nothing goes in the wrong box.
03
Use the result
Power, efficiency, steam rate and exhaust condition. Save it, reopen it, print it for the file.
PRICING
Start free. Upgrade when it earns its keep.
Monthly
Yearly 2 Months Free
Most popular
Trial
Free for 14 days. No card needed.
Pro
For engineers who size and check turbines every
Plant
The whole steam side of the plant, boiler to busbar.


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Start free. Upgrade when it earns its keep.
Monthly
Yearly 2 Months Free
Most popular
Trial
Free for 14 days. No card needed.
Free
14 days, then choose a plan
Start free trial
Turbine power calculator
All pressure, temperature and flow units
Live turbine diagram
5 saved calculations
Pro
For engineers who size and check turbines every week.
₹1,249 / month
₹14,990 billed yearly
Choose Pro
Power, efficiency and steam flow calculators
Advanced mode: stage, gearbox and generator losses
Exhaust dryness and expansion chart
200 saved calculations
Printable reports
Plant
The whole steam side of the plant, boiler to busbar.
₹4,166 / month
₹49,990 billed yearly
Choose Plant
Everything in Pro
Boiler steam generation with fuel blending
Unlimited saved calculations
Priority support
Prices exclude GST. Payments are in test mode during the preview: no money is taken.


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FAQ
Questions engineers ask first.
Which steam properties does it use?
Does it handle wet exhaust?
Which turbines is it for?
Can I use gauge pressure?
Is the payment real?
How many devices can I sign in on?
Run your first calculation in
under a minute.
Free for 14 days. No card, no steam tables, no spreadsheet.
Create your account

RotorCalc

Welcome back

Sign in to your calculators.

Work email

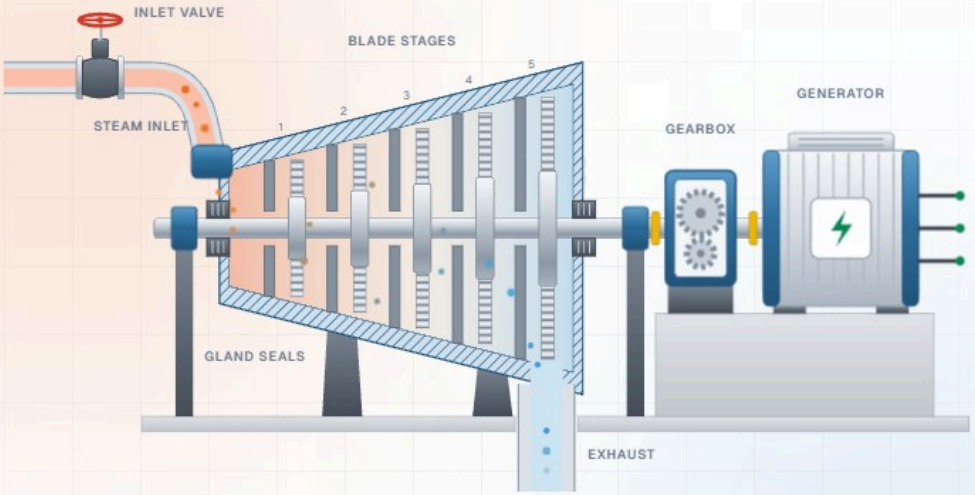
Password

Show

Forgot password?

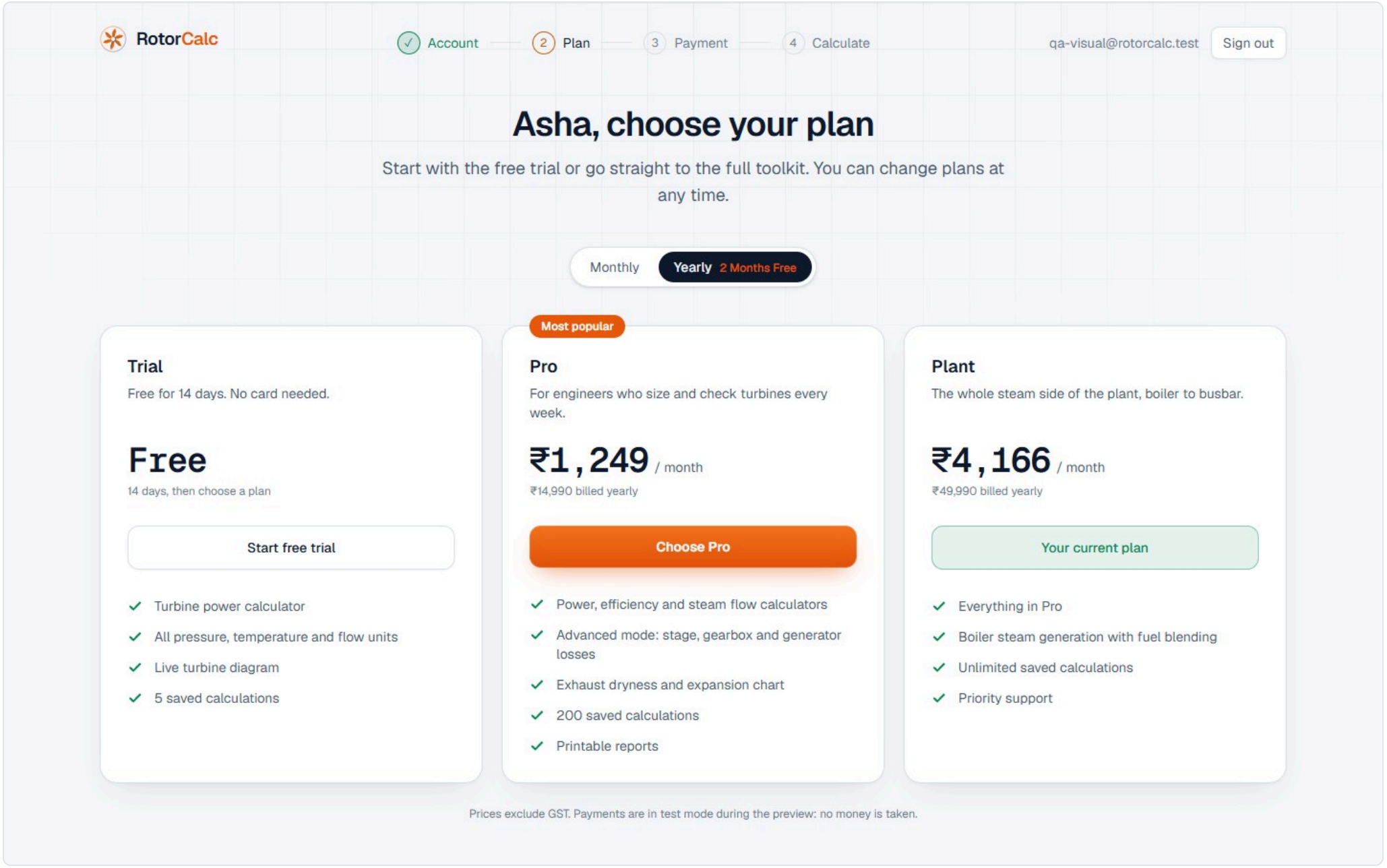
Sign in

New here? Create an account

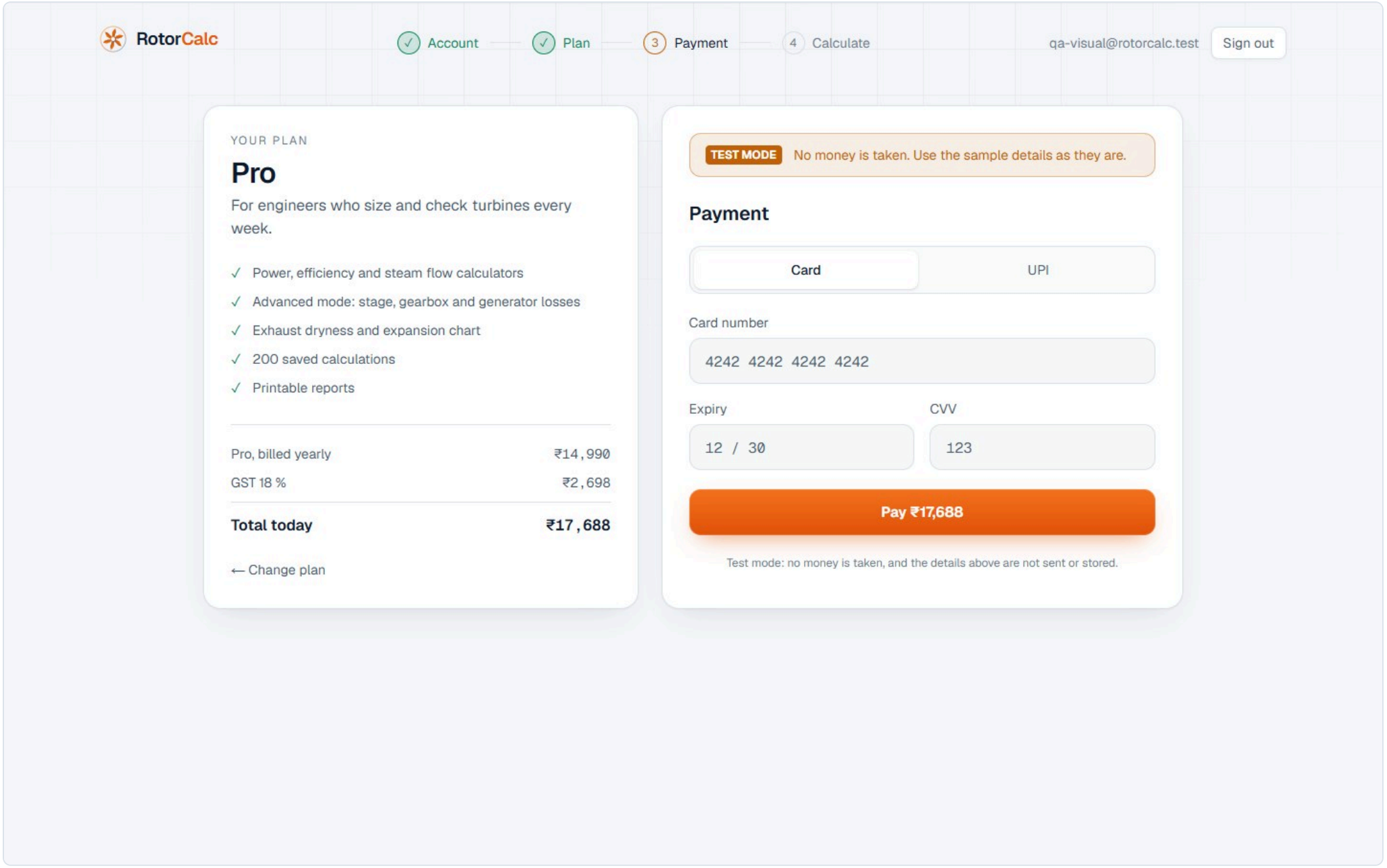


From gauge readings to a number you can defend.

Verified IAPWS-IF97 steam properties. Wet exhaust handled exactly. Every input shown on the machine.


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Account
2 Plan
3 Payment
4 Calculate
qa-visual@rotorcalc.test
Sign out
Asha, choose your plan
Start with the free trial or go straight to the full toolkit. You can change plans at any time.
Monthly
Yearly 2 Months Free
Most popular
Trial
Free for 14 days. No card needed.
Free
14 days, then choose a plan
Start free trial
Turbine power calculator
All pressure, temperature and flow units
Live turbine diagram
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Pro
For engineers who size and check turbines every week.
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Your current plan
Everything in Pro
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Unlimited saved calculations
Priority support
Prices exclude GST. Payments are in test mode during the preview: no money is taken.

Checkout (test mode) 12 / 21



RotorCalc | Power output | Efficiency | Steam flow | Boiler | Saved | Plant plan | AV | Sign out

Power output

Generator output from steam flow, inlet conditions and back pressure.

Basic | Advanced

1 Steam inlet

What arrives at the turbine stop valve

Steam flow: 50 TPH

Inlet pressure: 66 kg/cm²(a)

Inlet temperature: 485 °C

2 Exhaust

Where the expansion ends

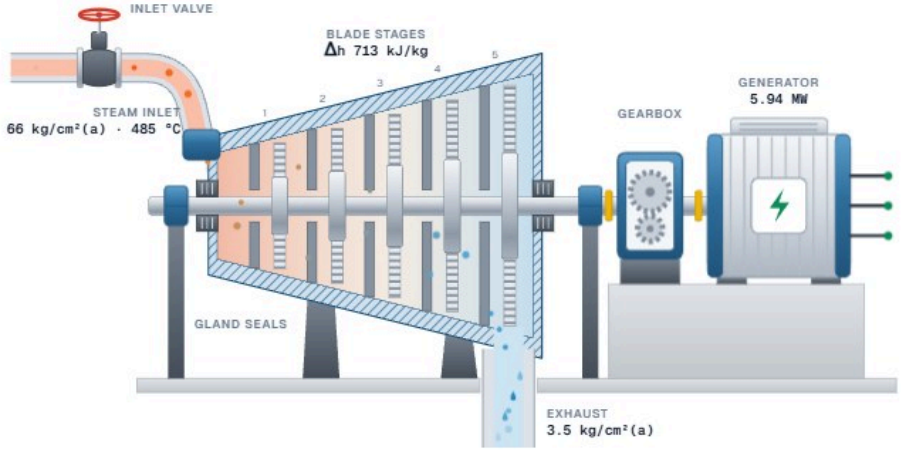
Back pressure: 3.5 kg/cm²(a)

3 Machine

One overall efficiency, steam to generator terminals

Turbine type: Impulse (typical 60 %)

Overall efficiency: 60 %



GENERATOR OUTPUT

5.939 MW

5,939 kW · 8,075 hp

Isentropic drop	Steam rate	Overall efficiency	Ideal exhaust
712.7 kJ/kg	8.42 kg/kWh	60.0 %	97.1 % dry

⚠ The ideal exhaust is wet steam at 97.1 % dryness.

Where the power goes

Power output, advanced

RotorCalc | Power output | Efficiency | Steam flow | Boiler | Saved | Plant plan | AV | Sign out

Inlet temperature: 485 °C

Inlet valve pressure drop: 3 %

Typical 2 to 5 %

2 Exhaust
Where the expansion ends

Back pressure: 3.5 kg/cm²(a)

Gland leakage: 0.5 %

Share of steam flow that bypasses the blades

3 Machine
Each loss on its own

Isentropic efficiency: 70 %

Blades and nozzles

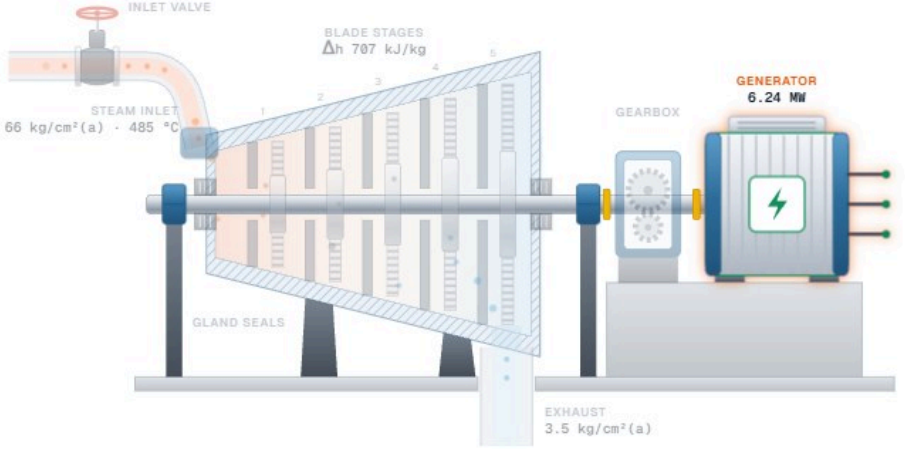
Mechanical efficiency: 98 %

Bearings and seals

Gearbox efficiency: 97 %

100 if direct drive

Generator efficiency: 96 %



GENERATOR OUTPUT

6.243 MW

6,243 kW · 8,489 hp

Isentropic drop	Steam rate	Overall efficiency	Exhaust
707.2 kJ/kg	8.01 kg/kWh	63.9 %	210.9 °C

Where the power goes

Ideal steam power — flow × isentropic drop	9,773 kW	100.0 %
Given to the rotor — after blade losses	6,841 kW	70.0 %
At the coupling — after bearings	6,705 kW	68.6 %
Generator output — after gearbox and generator	6,243 kW	63.9 %

RotorCalc | Power output | Efficiency | Steam flow | Boiler | Saved | Plant plan | AV | Sign out

Efficiency

Isentropic and overall efficiency from the readings you already log.

1 Steam inlet

Readings at the turbine stop valve

Steam flow: 50 TPH

Inlet pressure: 66 kg/cm²(a)

Inlet temperature: 485 °C

2 Exhaust

Readings at the turbine outlet

Exhaust pressure: 3.5 kg/cm²(a)

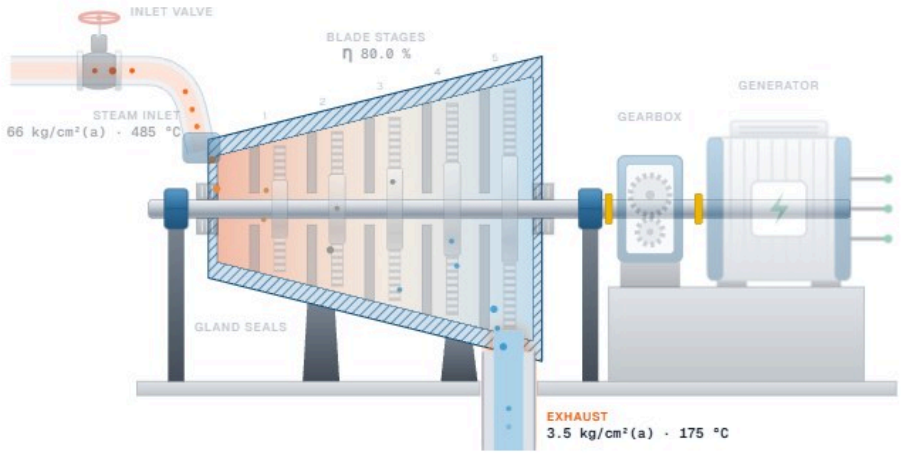
Exhaust temperature: 175 °C

Must be above the boiling point at exhaust pressure

3 Generator

Optional: adds overall and drive-train efficiency

Measured output: kW



ISENTROPIC EFFICIENCY

80.0 %

569.8 of 712.7 kJ/kg captured by the blades

Actual drop	Ideal drop	Overall efficiency	Exhaust superheat
569.8 kJ/kg	712.7 kJ/kg	-	36.8 °C

Where the power goes

Ideal steam power — flow × isentropic drop — 9,899 kW 100.0 %

Given to the rotor — flow × actual drop — 7,914 kW 80.0 %

RotorCalc | Power output | Efficiency | Steam flow | Boiler | Saved | Plant plan | AV | Sign out

Steam flow

The steam you need at the stop valve to reach a power target.

1 Target

What the generator must deliver

Required output

5000 kW

2 Steam conditions

Available at the inlet, and the back pressure

Inlet pressure

66 kg/cm²(a)

Inlet temperature

485 °C

Back pressure

3.5 kg/cm²(a)

3 Machine

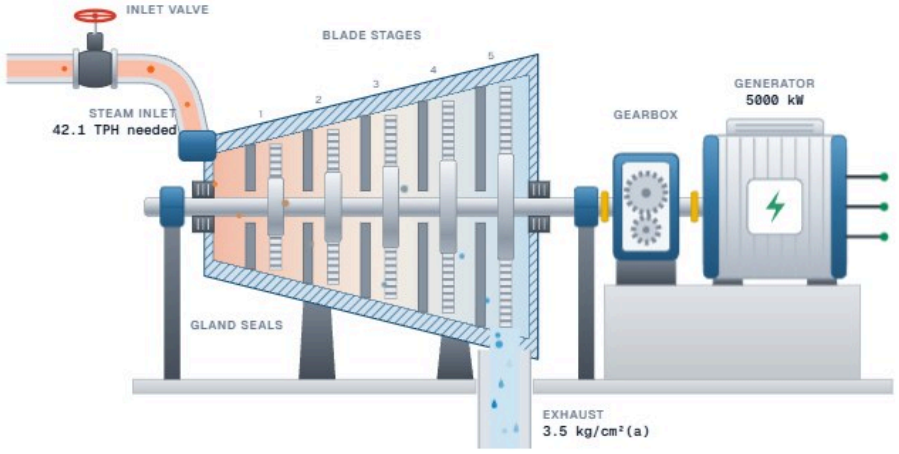
Overall efficiency, steam to generator terminals

Turbine type

Impulse (typical 60 %)

Overall efficiency

60 %



STEAM FLOW NEEDED

42.09 TPH

11.693 kg/s · 42,094 kg/h

Isentropic drop	Steam rate	Ideal steam power	Ideal exhaust
712.7 kJ/kg	8.42 kg/kWh	8,333 kW	97.1 % dry

⚠ The ideal exhaust is wet steam at 97.1 % dryness.

Steam properties

Expansion on the h-s chart

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Boiler

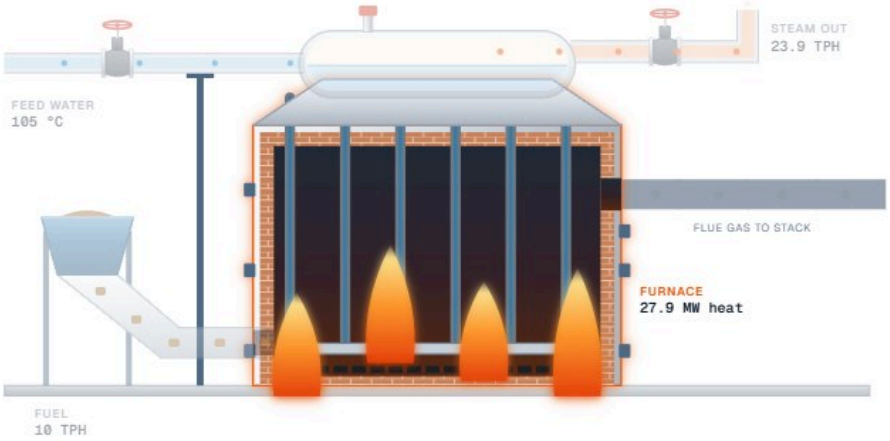
Steam generation from a fuel mix, on calorific value and boiler efficiency.

1 Fuel mix

Share of each fuel by weight, and its gross calorific value

Fuel	Share %	CV kJ/kg
Bagasse	100	10032
Rice husk	0	13376
Paddy straw	0	13376
Wood chips	0	12560
Indian coal	0	16750

Total share 100.0 %



2 Firing

How much fuel, and how well the boiler uses it

Fuel firing rate: 10 TPH

Boiler efficiency: 70 %

On gross calorific value

3 Steam and feed water

At the boiler outlet and the economiser inlet

STEAM GENERATION

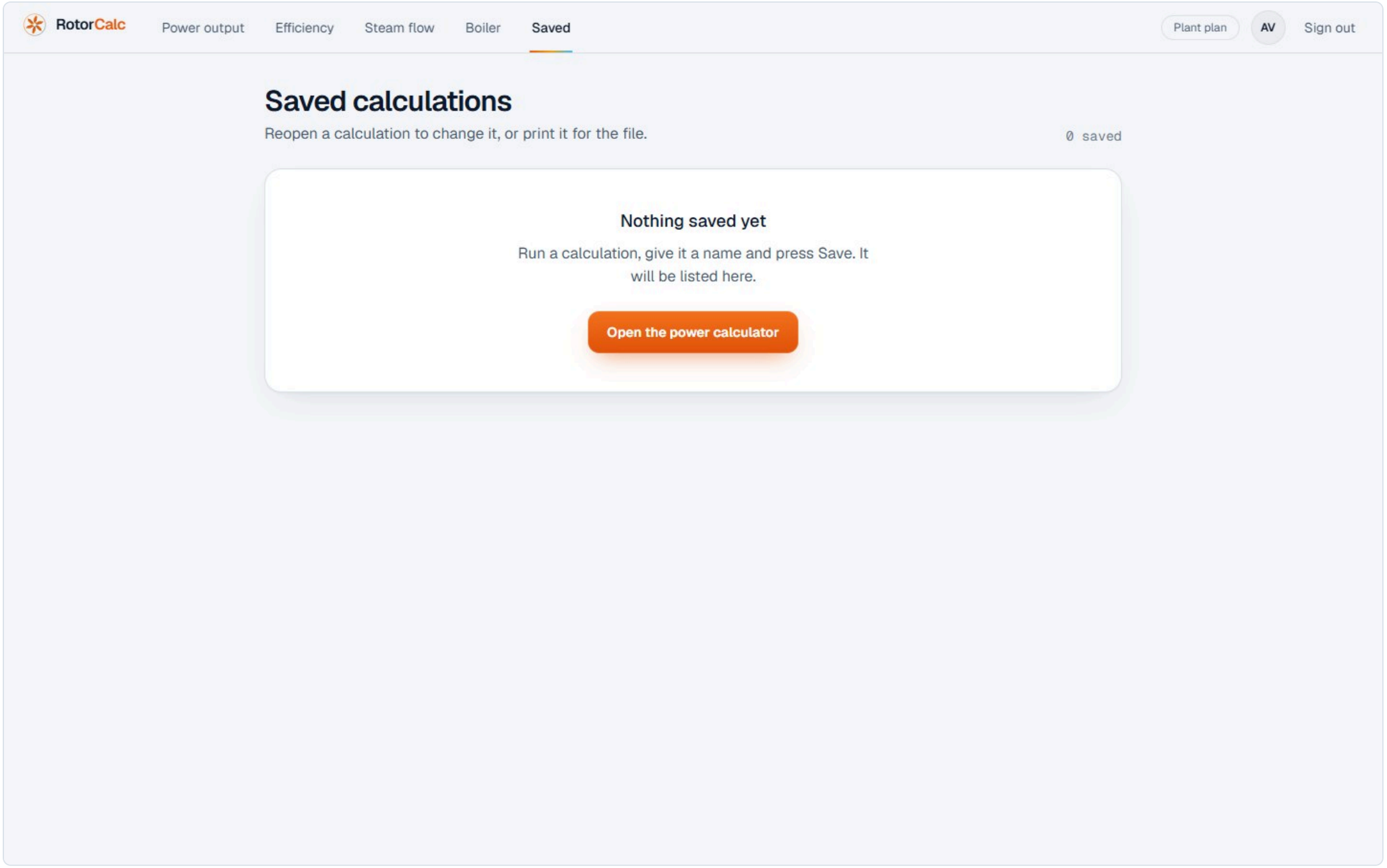
23.94 TPH

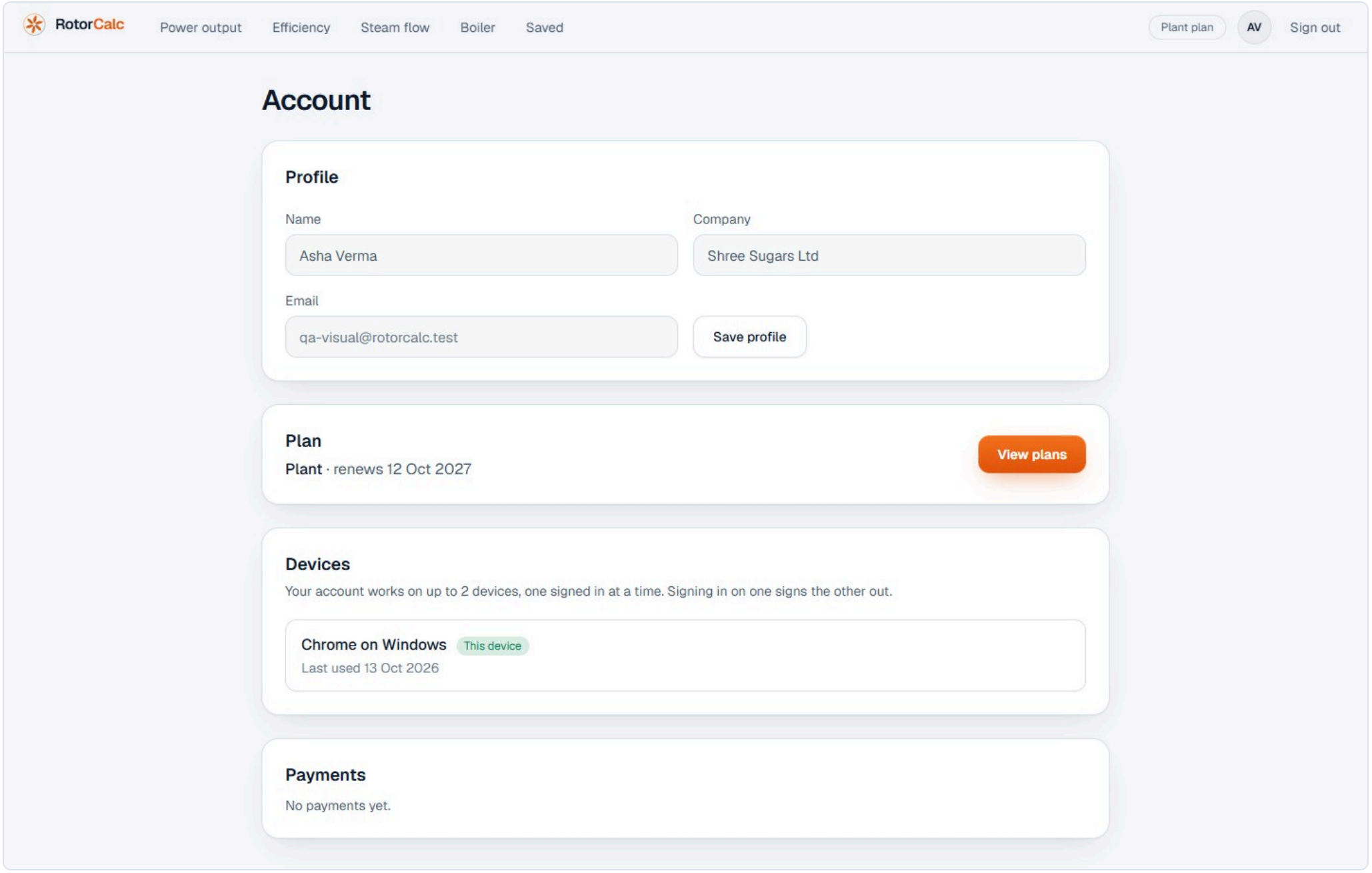
6.649 kg/s · 2.39 kg steam per kg fuel

Heat input	Heat to steam	Blended CV	Steam to fuel
27.87 MW	19.51 MW	10,032 kJ/kg	2.39 kg/kg

Steam and water properties

Steam enthalpy at boiler exit	3,378.8 kJ/kg
Feed water enthalpy	445.1 kJ/kg


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Power output
Efficiency
Steam flow
Boiler
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Plant plan
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Saved calculations
Reopen a calculation to change it, or print it for the file.
0 saved
Nothing saved yet
Run a calculation, give it a name and press Save. It will be listed here.
Open the power calculator

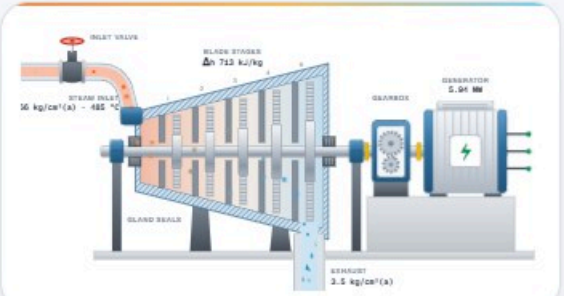

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Power output
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Steam flow
Boiler
Saved
Plant plan
AV
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Account
Profile
Name
Asha Verma
Company
Shree Sugars Ltd
Email
qa-visual@rotorcalc.test
Save profile
Plan
Plant · renews 12 Oct 2027
View plans
Devices
Your account works on up to 2 devices, one signed in at a time. Signing in on one signs the other out.
Chrome on Windows
This device
Last used 13 Oct 2026
Payments
No payments yet.

RotorCalc Power outp AV Sign out

Power output

Generator output from steam flow, inlet conditions and back pressure.

Basic Advanced



GENERATOR OUTPUT

5.939 MW

5,939 kW · 8,075 hp

Isentropic drop 712.7 kJ/kg	Steam rate 8.42 kg/kWh
Overall efficiency 60.0 %	Ideal exhaust 97.1 % dry

⚠ The ideal exhaust is wet steam at 97.1 % dryness.

Landing page on a phone 21 / 21

